

Sunmay Padiyar

sunmaypadiyar.me | linkedin.com/in/sunmay-padiyar | github.com/S-Padiyar | U.S. Citizen

EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Science; Junior standing by credit hours

Aug. 2026 – May 2029

- Coursework: Data Structures & Algorithms, Computer Organization & Programming, Computing & Society, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Object-Oriented Programming

Broad Run High School

Ashburn, VA

Advanced Diploma; GPA: 4.88 weighted; SAT: 1560 (800 Math); 20+ AP/DE Courses

Aug. 2022 – June 2026

EXPERIENCE

Nova Cloud LLC

Remote

Software Engineering Intern

Mar. 2026 – Present

- Work alongside senior developers to build and ship a client-facing cloud management platform using Vue.js, Node.js, Express, Redis, and DynamoDB.
- Integrate data from 40+ AWS services, including EC2, S3, RDS, Lambda, and Cost Explorer, to help clients monitor usage and identify cost-optimization opportunities across multiple accounts.
- Build a mock backend for ~100 API endpoints, allowing the team to develop, test, and demonstrate to clients without accessing live AWS accounts or credentials.
- Create Selenium end-to-end tests covering 50+ application routes, including authentication, permissions, navigation, and cost-management features.
- Use Docker, Selenium Grid, and Healenium to support automated testing while integrating AWS Bedrock for AI-powered cloud analysis and recommendations.

George Mason University

Fairfax, VA

AI/ML Research Intern

June 2025 – Mar. 2026

- Conducted prompt engineering research under Dr. Mihai Boicu, evaluating performance of 4 prompt strategies across 90 advanced competitive programming problems (rated 1600-2400).
- Developed an automated workflow that generated Java solutions and submitted them unedited to the official CodeForces grader, including a 2-model prompt-chaining process with up to 5 critique-and-revision rounds.
- Led the research project as first author and presented the peer-reviewed IEEE paper at the 2025 MIT URTC and GMU Research Symposium, overseeing the methodology, experimentation, paper writing, and statistical analysis (χ^2 , McNemar's).

FTC Team #23541 CyberSages

Ashburn, VA

Founder & Captain

June 2023 – Jan. 2026

- Founded a 10-member FTC robotics team across 3 seasons, secured \$4,000+ in grants and sponsorships, coordinated software, mechanical, and electrical development, with the team advancing to the State Championship and earning the Think Award.
- Designed a 4-wheel mecanum-drive robot with a custom-cut metal frame and resin-printed parts; wired and integrated the motors, servos, sensors, control hubs, and power system for use across 30+ official matches.
- Built modular Java autonomous software using Road Runner and PID/PIDF control, tuning trajectories and mechanism gains through telemetry and field testing to increase consistency by ~30% and decrease runtime by ~40%.

PROJECTS

CodeForces AI Evaluation Pipeline

2025 – 2026

Python, Selenium, undetected ChromeDriver, OpenAI API, Codeforces API, Pandas

- Built a Python framework to collect 90+ random Codeforces problems, retrieve ratings and tags through the Codeforces API, cache problem statements, and organize reproducible evaluation datasets.
- Automated the full evaluation process using the OpenAI API, Selenium, and undetected ChromeDriver, generating Java solutions, submitting them unedited to the official Codeforces grader, and recording each verdict.
- Compared 4 prompting strategies with Structured Chain-of-Thought increasing acceptance from 44.44% to 56.67%; Implemented Python-based χ^2 and McNemar's tests used to measure significance.

FTC Autonomous Navigation & Control System

2023 – 2026

Java, FTC SDK, Road Runner, PID/PIDF, three-wheel dead-wheel odometry

- Programmed 4-wheel mecanum drive and autonomous movement using dead-wheel odometry, including a position-hold feature that moves the robot back to its original pose despite being pushed.
- Created 1-button goal alignment using heading data; tuned the launcher's power with PIDF control for fast ball-launching.
- Wrote reusable, modular Java controls for motors and sensors across 3 FTC seasons, including code for a motorized retractable arm; utilized trajectory visualization software to plan autonomous paths before field testing.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Tools: React, Next.js, Vue.js, Node.js, Flask, REST APIs, AWS, Git/GitHub, Linux, Selenium

Additional: OpenAI API, Pandas, FTC SDK, Road Runner, PID/PIDF control, dead-wheel odometry